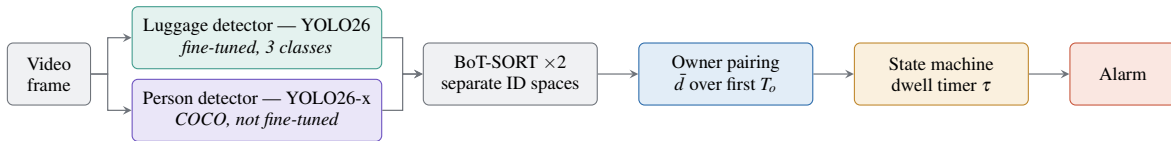
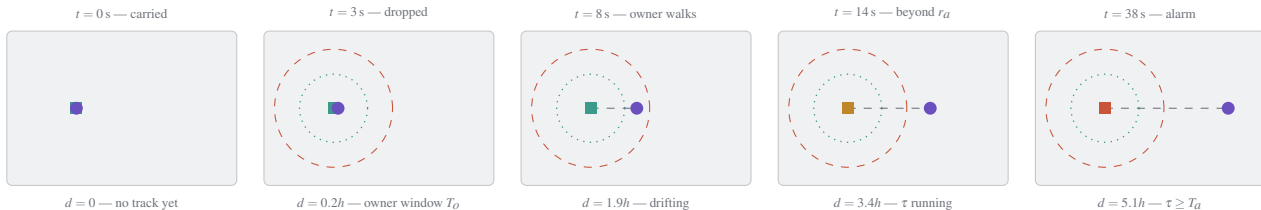


(a) Per-frame pipeline



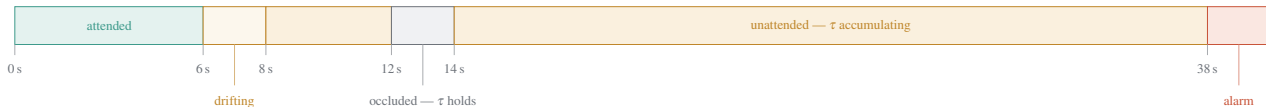
(b) Worked example — one luggage track, $r_o = 1.5h$, $r_a = 2.5h$, $T_a = 30$ s

radii in person heights h , so they scale with depth



(c) State and dwell timer τ on a true time axis

the (b) snapshots are evenly spaced; here time is to scale



Guard 1. The owner is the person minimising mean distance over the bag's first T_o seconds — not the nearest person in any single frame.

Guard 2. τ advances only while the bag track is alive, so the occlusion at $t = 12$ – 14 s is not counted as a departure.